Troubleshooting Tools

1. Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.

ANS :

```
C:\Users\HARSH>ipconfig

Windows IP Configuration


Wireless LAN adapter Local Area Connection* 4:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 5:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::54ba:4a02:fe34:ed65%13
   IPv4 Address. . . . . . . . . . . : 192.168.247.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::afab:7e61:830d:9cb9%4
   IPv4 Address. . . . . . . . . . . : 192.168.101.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::387c:745d:bc9c:c1d0%21
   IPv4 Address. . . . . . . . . . . : 192.168.0.101
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.0.1

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

2. Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.

ANS :

```
C:\Users\HARSH>ping www.google.com

Pinging www.google.com [142.251.155.119] with 32 bytes of data:
Reply from 142.251.155.119: bytes=32 time=14ms TTL=116
Reply from 142.251.155.119: bytes=32 time=14ms TTL=116
Reply from 142.251.155.119: bytes=32 time=13ms TTL=116
Reply from 142.251.155.119: bytes=32 time=14ms TTL=116

Ping statistics for 142.251.155.119:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 13ms, Maximum = 14ms, Average = 13ms
```

3. Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.

ANS :

```
C:\Users\HARSH>tracert www.flipkart.com

Tracing route to flipkart.com [163.53.76.86]
over a maximum of 30 hops:

  1     1 ms     1 ms     2 ms  192.168.0.1
  2     8 ms     1 ms     1 ms  dsldevice.lan [192.168.1.254]
  3    10 ms     4 ms     2 ms  10.130.192.1
  4     *        *        *     Request timed out.
  5    12 ms    12 ms    12 ms  136.232.112.109
  6     *        *        *     Request timed out.
  7    30 ms    27 ms    28 ms  115.242.187.186
  8     *        *        *     Request timed out.
  9     *        *        *     Request timed out.
 10     *        *        *     Request timed out.
 11     *        *        *     Request timed out.
 12    34 ms    34 ms    33 ms  163.53.76.86

Trace complete.
```

4. Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.

ANS :

```
C:\Users\HARSH>nslookup wwspotify.com
Server:  UnKnown
Address:  192.168.0.1

Non-authoritative answer:
Name:    atc.spotify.map.fastly.net
Addresses:  2a04:4e42:19::810
          151.101.107.42
Aliases:  www.spotify.com
```

- **One-Line Explanation**

- nslookup verifies that a domain name is correctly translated into its IP address by the DNS server.

5. A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check. Hint: Think about checking both network connectivity and DNS resolution.

ANS :

Step	Command	Purpose
1	ipconfig	Check that the computer has a valid IP address and default gateway.
2	nslookup www.zomato.com	Verify that DNS is resolving the website name to an IP address.
3	ping www.zomato.com	Test whether the website or its server is reachable from the computer.

- Explanation

1. ipconfig checks whether the network configuration is correct.
2. nslookup confirms that DNS is working properly.
3. ping tests connectivity and identifies whether the destination server can be reached.